CHA-3.C.1 The derivative can be used to solve problems involving rates of change in applied contexts.

4-3 Rates in Context

Previously in Unit 4, we have looked at units of derivatives in context, interpreting derivatives in context, and using derivatives to solve problems involving straight-line motion. Using these concepts together, we can apply our knowledge of derivatives in other contexts.

Increasing or Decreasing	To know if something is **increasing** or **decreasing** (or how the rate is **changing**), check the **sign of its derivative**!

Height is *increasing* if $[\textbf{height}]' > 0$

Velocity is *decreasing* if $[\textbf{velocity}]' < 0$

Mr. Stark's IQ is *increasing* if $[\textbf{Mr. Stark's IQ}]' > 0$

What if the function is a rate of change?

If $f(x)$ is the bunny population after x years, then what is $f'(x)$?

Rate of change of **<u>the bunny population</u>** per **<u>year</u>**

If $f(x)$ is the rate at which a bunny population changes (bunnies per year), then what is $f'(x)$?

Rate of change of **<u>the rate at which the bunny population changes per year</u>** per **<u>year.</u>**

Example 1: Speedville High School opens its doors for students at 8AM each morning. On one particular morning, the number of students inside Speedville High School can be modeled by the function $P(t) = 25\ln(3t+1)$ where $0 \le t \le 8$, and t is the number of minutes since the doors open at 8AM. Find the rate that the number of students inside Speedville HS is changing at 8:03AM.

$P'(3) = ?$

$P(t) = 25\ln(3t+1)$

$P'(t) = 25 \cdot \frac{1}{3t+1} \cdot 3 = \frac{75}{3t+1}$

$P'(3) = \frac{75}{9+1} = \frac{75}{10} = 7.5$

$\frac{\text{Units of P}}{\text{Units of t}}$

The number of students inside Speedville HS is increasing at a rate of 7.5 students/min at 8:03 AM.

Example 2: A rumor is spreading around school that Mr. Young once placed 3rd in the World's Strongest Man contest. The rumor is spreading at the rate $\boxed{\frac{dR}{dt} = \frac{1}{4}R(500 - R)}$ where $R(t)$ is the total number of students that have heard the rumor and t is the number of hours since the rumor began. Is the rumor spreading faster when 200 people have heard it or when 400 people have heard it? Explain your reasoning.

$$\left.\frac{dR}{dt}\right|_{R=200} = \frac{1}{4}(200)(500-200) = 50(300) = 15000 \text{ Stud/hr}$$

$$\left.\frac{dR}{dt}\right|_{R=400} = \frac{1}{4}(400)(500-400) = 100(100) = 10000 \text{ Stud/hr}$$

The rumor is spreading faster when 200 people have heard it, because $\left.\frac{dR}{dt}\right|_{R=200} > \left.\frac{dR}{dt}\right|_{R=400}$

Example 3 (Calculator Active): For $0 \le s \le 5000$, the total amount a local college spends on printing is related to the number of students enrolled at the school and is modeled by $C(s) = 70 - 32\cos\left(\frac{s}{2000}\right)$ where s is the number of students enrolled at the college and C is the total cost of printing, in thousands of dollars. At what rate is the total cost of printing changing, in dollars per student, when 2500 students are enrolled at the college?

$C'(2500) = 0.015183$

The total cost of printing increasing at a rate of 0.015 thousands of dollars/student (or \$15.183/student) when 2500 students are enrolled.

Example 4: For $0 \le t \le 10$ hours, the rate that cars enter a toll road, in cars per hour, is modeled by $C(t) = 400 - 5(t-2)^2$. During this time, cars exit the toll road at a constant rate of 200 cars per hour. Is the number of cars on the toll road increasing or decreasing at time $t = 9$ hours?

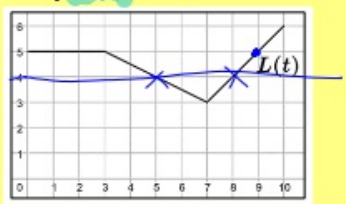
Rate cars enter at $t=9$: $C(9) = 400 - 5(7)^2 = 155$ cars/hr

Rate cars leave at $t=9$: 200 cars/hr

Rate of Cars on Road = Rate enter − Rate leave

At t = 9, the cars enter at a rate of 155 cars/hour and exit at a rate of 200 cars/hour, so the number of cars on the toll road is decreasing at a rate of 45 cars/hour.

Example 5: At the local library, visitors check out and return books throughout the day. The rate that visitors check out books, in books per minute, is modeled by $L(t)$ where t is measured in minutes. During this time, the total amount of books that have been returned to the library is modeled by $R(t) = 4t$, where t is measured in minutes.



(a) At what rate are books being checked out by visitors at time $t = 9$ minutes?

$L(9) = 5$ books/min

(b) Is the amount of books at the library increasing or decreasing at time $t = 6$ minutes?

Rate books are returned : $R'(t) = 4$

Rate books are checked out : $L(t)$

$t = 6$

$R'(6) = 4$ books/min

$L(6) = 3.5$ books/min

At $t = 6$, books are being returned at a rate of 4 books/min and are being checked out at a rate of 3.5 books/min, so the amount of books in the library is increasing at a rate of 0.5 books/min

(c) At what time(s) is the rate that books are checked out equal to the rate that books are being returned?

$L(t) = R'(t)$

$L(t) = 4$

$t = 5$ min $t = 8$ min